Exploratory Data Analysis & Predictive Modeling

Executive Summary Report: TikTok Verified Status Classification

Overview

The main goal of this stage of the project is to build a Logistic Regression model to predict account verification status (verified vs. unverified) based on video engagement metrics and content characteristics. By understanding which behavioral and content features drive verification status, TikTok can optimize moderation workflows and refine account verification rules.

Problem & Data Structure

During data prep and feature evaluation, critical structural insights were identified:Content Type Disparity (Claims vs. Opinions): Opinion videos are significantly more likely to be authored by verified accounts than claim videos. Out of all verified user posts, the vast majority consist of opinions rather than claims.

- **Multicollinearity & Feature Redundancy**: High pairwise correlation exists among video engagement variables (`video_view_count`, `video_like_count`, `video_share_count`, `video_download_count`, `video_comment_count`). To avoid multicollinearity in logistic regression, engagement features were evaluated or transformed carefully.
- **Target Imbalance & Encoding:** The target variable `verified_status` was binary encoded (1 = verified, 0 = unverified). Verified accounts represent a smaller proportion of total posts compared to unverified creators.

Key Insights

Feature	Coefficient
video_comment_count	-0.000608
video_download_count	-0.000041
author_ban_status_banned	-0.000013
author_ban_status_under review	-0.000005
video_view_count	-0.000002
video_share_count	0.000007
claim_status_opinion	0.000384
video_duration_sec	0.008870

Strategic model findings and coefficient interpretations:

- **Claim Status as the Dominant Predictor:** The categorical feature `claim_status` (specifically `claim_status_opinion`) holds the strongest statistically significant coefficient in predicting account verification. Holding engagement constant, posting an opinion increases the odds of an account being verified by several orders of magnitude compared to posting a claim.

- **Minimal Engagement Metric Impact:** While view count and share count are slightly higher for unverified accounts posting claims, engagement metrics alone show near-zero log-odds coefficients when controlling for claim status.

- **Classification Threshold Optimization:** Adjusting probability decision thresholds allowed balancing precision and recall to align with TikTok moderation policies.

	precision	recall	f1-score	support
0.0	0.76	0.47	0.58	5353
1.0	0.62	0.85	0.72	5378
accuracy			0.66	10731
macro avg	0.69	0.66	0.65	10731
weighted avg	0.69	0.66	0.65	10731

Next Steps & Recommendations

Strategic roadmap for deployment and platform policy:

- **Moderation Queue Prioritization:** Use claim classification as an upstream filter for account verification reviews, routing creators who post original opinions and high engagement through expedited verification tracks.
- **Misinformation & Claim Monitoring:** Unverified accounts posting claims with viral engagement should be flagged for safety review rather than automatic verification.
- **Advanced Non-Linear Modeling:** Proceed to tree-based ensemble models (Random Forest, XGBoost) in the next project phase to capture non-linear interactions between engagement spikes and author status.